SlipSurfaceIQ LE

# DXF Import Guide

Bring slope cross-sections from CAD into SlipSurfaceIQ LE

Version 2026.6.0

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# 01 Overview

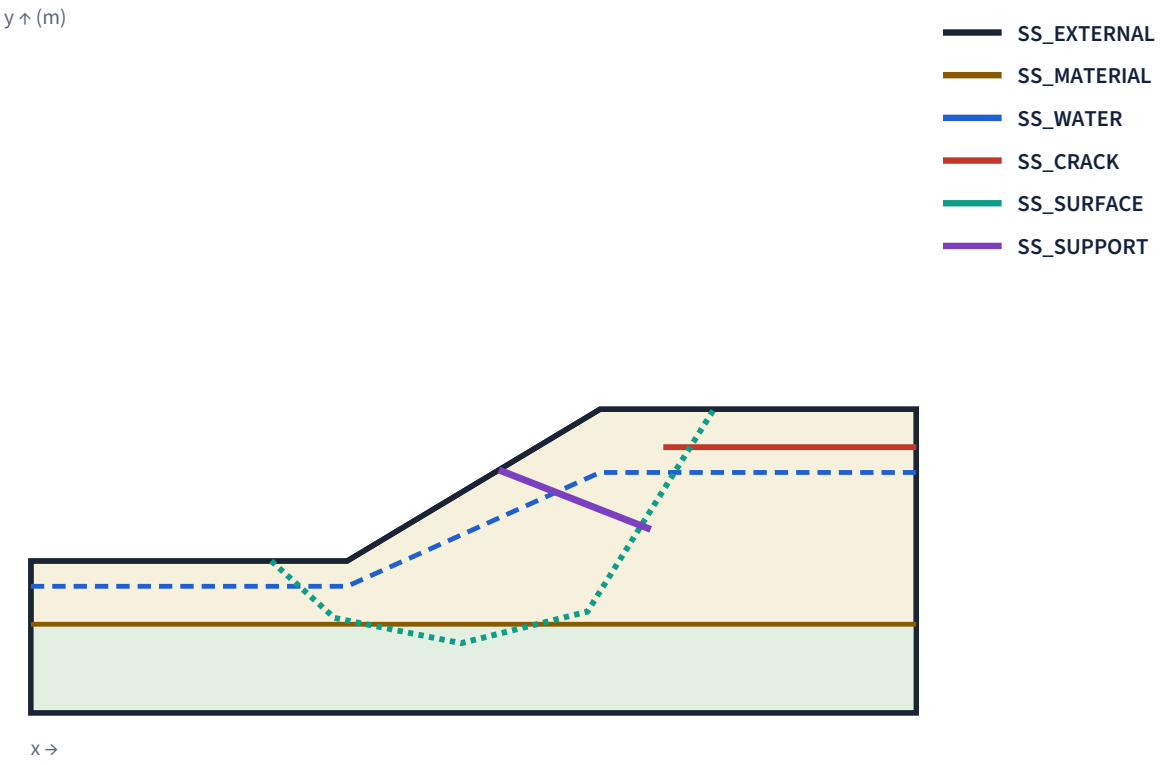
File → Import DXF reads the 2D geometry of a slope from a CAD drawing. Each kind of line is recognised by its layer name.

• **Geometry only.** The external boundary, material boundaries, water table, tension crack and a user slip surface are read. Materials, loads, supports and settings are not part of DXF.

• **Replaces the open model.** Importing creates a new model with one default material (Soil 1) and the file name as title. **Ctrl+Z** brings the previous model back.

• **Checked on import.** The geometry is built immediately; any problem is reported and nothing changes. Skipped entities and layers are listed in the message after a successful import.

• **File → Export DXF** writes the same layers, so a model can go to CAD and back.




The layers of a slope model. The same model is in the template drawing **docs/DXF_Layer_Template.dxf**.

# 02 Layer reference

Layer names are not case-sensitive (ss_water = SS_WATER).

| Layer | Entities | Becomes | Rules |
|---|---|---|---|
| `SS_EXTERNAL` | exactly 1, closed | External boundary | Required. One closed, non-self-intersecting polyline. |
| `0` | 1 closed | External boundary | Used when SS_EXTERNAL is not used: a closed polyline on layer 0 counts as the outline. Open entities on layer 0 are skipped. |
| `SS_MATERIAL` | any number | Material boundaries | Open lines that split the domain into regions; ends must touch the outline (see chapter 05). |
| `SS_WATER` | at most 1 | Water table | Left to right with increasing x; extended horizontally to both model ends. |
| `SS_CRACK` | at most 1 | Tension crack line | Its depth below the ground surface is the crack depth. |
| `SS_SURFACE` | at most 1 | User slip surface | Stored as the user-defined surface; select it in Surface Options. |
| `SS_SUPPORT` | export only | — | Written on export (one line per support); skipped on import. |
| `any other` | — | skipped | Listed in the import message. |

**Start from the template**
Open **docs/DXF_Layer_Template.dxf** in your CAD program: it already has every layer and one entity on each. Draw over it, delete what you do not need, and save as DXF.

# 03 Entities, curves and units

What is read from the drawing and how it is converted to metres.

## Entities

| Entity | Read as |
|---|---|
| `LINE` | Two-point line. |
| `LWPOLYLINE` | Polyline; arc segments (bulges) are converted to straight pieces. |
| `POLYLINE` | 2D polyline (same as LWPOLYLINE). |
| `ARC` | Converted to straight pieces. |
| `SPLINE` | Converted to straight pieces. |
| `CIRCLE, ELLIPSE, TEXT, MTEXT, INSERT (blocks), HATCH, DIMENSION, POINT, …` | Skipped and listed in the import message. |

Curves are replaced by straight segments that stay within **1 cm** of the true curve. Blocks are not exploded: explode them in CAD first.

## Units

The drawing unit ($INSUNITS, set with the UNITS command in most CAD programs) is converted to metres. A drawing without units is read as metres, with a warning.

| Drawing unit | Factor to metres | Example: 12 500 drawing units |
|---|---|---|
| `Millimetres` | 0.001 | 12.5 m |
| `Centimetres` | 0.01 | 125 m |
| `Metres` | 1 | 12 500 m |
| `Feet` | 0.3048 | 3 810 m |
| `Inches` | 0.0254 | 317.5 m |
| `Unitless` | 1 (warning) | 12 500 m |

**Coordinates**
x is horizontal and y is vertical (elevation), both in the drawing plane at Z = 0. A cross-section drawn in a rotated UCS or at an elevation Z must be moved to the world XY plane first. Geometry with Z ≠ 0 is refused.

# 04 Preparing a drawing in CAD

A checklist that avoids nearly every import error.

| # | Step | Typical commands |
|---|---|---|
| 1 | Copy the cross-section into a new drawing. | WBLOCK, COPYCLIP |
| 2 | Set the drawing units (preferably metres). | UNITS, INSUNITS |
| 3 | Explode blocks and groups. | EXPLODE |
| 4 | Move everything to Z = 0 in the world XY plane. | FLATTEN, UCS World |
| 5 | Join the outline into ONE polyline and close it. | JOIN, PEDIT → Join / Close |
| 6 | Put each line on its layer (SS_EXTERNAL, SS_MATERIAL, …). | LAYER, Properties |
| 7 | Join the water table into one polyline running left to right. | JOIN, REVERSE |
| 8 | Extend material boundaries until they meet the outline. | EXTEND, object snaps |
| 9 | Delete hatches, text and dimensions (or keep them on other layers). | ERASE |
| 10 | Save as DXF (any version, e.g. R2010). | SAVEAS → DXF |

# 05 Layer details

## External boundary — SS_EXTERNAL

One closed outline of the whole model: ground surface on top, the sides and the base. The vertex order does not matter. Separate LINE entities are not joined automatically; the import stops with “SS_EXTERNAL must be a closed polyline.”

## Material boundaries — SS_MATERIAL

Each open line divides the region it crosses. Line ends lying close to the outline or to another boundary are snapped onto it. After the import, assign materials with **Properties → Assign Material**; until then every region uses Soil 1.

> **Ends must reach the outline**
> Ends are snapped within **0.2 % of the larger model dimension** (12 cm for a 60 m wide model). A boundary that stops farther away is kept but does not split the region; the import message then says “Material boundary N does not divide a region”, and the Modeler status bar repeats it. Extend the lines onto the outline in CAD.

## Water table — SS_WATER

One polyline with increasing x (left to right, no back-steps). It is extended horizontally from its first and last points to the model ends. After import the groundwater method is *water table*; per-material Hu is set in the material dialog.

## Tension crack — SS_CRACK

One line or polyline above or on the ground; at least two points with different x. The crack depth at each x is its distance below the ground surface. Set the water filling in **Project Settings → Groundwater**.

### User slip surface — SS_SURFACE

One polyline, arc or spline. It is stored as the user-defined surface, but the surface type is not switched: choose **Surfaces → Surface Options → User-defined** to analyse it.

## 06 After the import

Complete the model in the Modeler.

**1.** Read the import message: it lists skipped entities and layers.

**2.** Define materials (**Properties → Define Materials**) and assign them to the regions.

**3.** Add loads, supports and facings, and set seismic coefficients if needed.

**4.** Choose the slip surface type and search options.

**5. Compute (F5)**, then save the model as **.slp.json** — that file keeps everything, DXF keeps only geometry.

## 07 Exporting to DXF

**File → Export DXF** writes an AutoCAD R2010 drawing in metres with the layers below. It needs an external boundary.

| Written | Not written |
|---|---|
| SS_EXTERNAL (closed), SS_MATERIAL, SS_WATER, SS_CRACK, SS_SURFACE, SS_SUPPORT (single supports) | Materials and assignments, loads, bolt patterns, facings, notes, settings and results |

Exported files import back unchanged (the supports layer is skipped on import), so DXF is a safe way to edit the geometry in CAD and return.

# 08 DXF from Python

The import and export functions can be used in a Python script (Tools → Python scripting) or any Python program.

import_in_a_script.py

```python
from slipsurface.exchange import import_dxf

project, warnings = import_dxf(TEMPLATE)   # path of a .dxf file
for line in warnings:
    print(line)
project.materials[0].c = 10   # continue the model as usual
project.materials[0].phi = 28
```

In the scripting editor write the file path instead of `TEMPLATE`, e.g. `r"C:\models\section.dxf"`.

write_a_dxf_with_ezdxf.py

```python
import ezdxf
from pathlib import Path
from slipsurface.exchange import import_dxf

doc = ezdxf.new("R2010")
doc.units = ezdxf.units.M
msp = doc.modelspace()
for layer in ("SS_EXTERNAL", "SS_MATERIAL", "SS_WATER"):
    doc.layers.add(layer)
outline = [(0, -10), (60, -10), (60, 10), (40, 10), (20, 0), (0, 0)]
msp.add_lwpolyline(outline, close=True, dxfattribs={"layer": "SS_EXTERNAL"})
msp.add_line((0, -4), (60, -4), dxfattribs={"layer": "SS_MATERIAL"})
msp.add_lwpolyline([(0, -2), (20, -2), (40, 5), (60, 5)],
                   dxfattribs={"layer": "SS_WATER"})
path = Path(TMP) / "section.dxf"
doc.saveas(path)
project, warnings = import_dxf(path)
print(len(project.build_geometry().regions), "regions")
```

Export works the same way: `from slipsurface.exchange import export_dxf` and `export_dxf(project, "section.dxf")`.

# 09 Messages and fixes

Import messages, word for word, and what to do.

| Message | Cause and fix |
|---|---|
| `DXF needs exactly one closed SS_EXTERNAL (or layer 0) boundary.` | No external boundary, or more than one closed polyline on SS_EXTERNAL / layer 0. Keep one closed outline; move other closed shapes to another layer. |
| `SS_EXTERNAL must be a closed polyline.` | The outline is open, or drawn as separate LINEs. Join it into one polyline and close it (JOIN / PEDIT → Close). |
| `Multiple entities on SS_WATER; join them before importing.` | SS_WATER, SS_CRACK or SS_SURFACE holds more than one entity. Join them into one polyline. |
| `Only planar XY geometry at Z=0 is supported.` | Some vertices have Z ≠ 0. Flatten the drawing (FLATTEN, or set Z to 0). |
| `The external boundary intersects itself (invalid polygon).` | The outline crosses itself. Fix the vertex order or remove the loop. |
| `The external boundary has zero area.` | The outline is degenerate (all points on a line). |
| `Water table points must be in increasing x order.` | The water table runs right to left or turns back. Reverse it (REVERSE) and remove back-steps. |
| `The tension crack line needs at least 2 points with different x values.` | The crack line is a single point or vertical. |

| Notice after a successful import | Meaning |
|---|---|
| `DXF has no units; coordinates interpreted as metres.` | Set the drawing units if the drawing is not in metres. |
| `Skipped entities/layers: …` | These entities were ignored (see chapter 03). |
| `Material boundary N does not divide a region (… m of it is loose): …` | Its ends do not reach the outline or another boundary; extend them (chapter 05). |
| `DXF transfers geometry only. Assign materials, loads and supports separately.` | Always shown; continue with chapter 06. |